moepager

Expert-Aware Page-Cache Management for Mixture-of-Experts LLM Inference on Memory-Constrained Linux Machines

Formal title: System and Method for Non-Invasive, Expert-Aware Page-Cache Management for Mixture-of-Experts Model Inference on Memory-Constrained Linux Hosts

Repository: https://github.com/shreshth006/moepager · team fork: https://github.com/AyushSinha2210/moepager (branch `ayush`)

1. Title, Abstract and Architecture

1.1 Abstract

Mixture-of-Experts (MoE) language models activate only a few "expert" sub-networks per layer for each token: 8 of 64 in OLMoE-1B-7B, 4 of 32 in gpt-oss-20b, 8 of 128 in Qwen3-30B-A3B. That makes them attractive for consumer machines, which run them with llama.cpp in `mmap` mode when the model is larger than free RAM, leaving the Linux page cache to decide which weights stay in memory.

We show that the kernel's generic behaviour fits this workload poorly:

- **Scattered units.** One expert's weights are three slices in three tensors hundreds of megabytes apart.
- **Read-around pollution.** On a page fault, the kernel reads a window around the faulting page: 4 MiB on our test machine's btrfs. That pulls unused neighbouring experts into the cache and multiplies SSD traffic 3.7–4.8×.
- **Repacking.** llama.cpp's default CPU weight repacking moves 77–100 % of expert bytes into anonymous memory, outside the page cache.

We present **moepager**, a Rust toolchain and an unprivileged user-space daemon that manage the shared page cache at the granularity of (layer, expert) units, without changing the inference engine's code or binary:

1. It parses the GGUF header into an exact expert map.
2. It observes expert misses black-box by polling sentinel pages with `mincore()`.
3. It answers each miss with **expert-completion readahead**: bulk `posix_fadvise(WILLNEED)` of the expert's remaining slices in 128 KiB chunks. The chunking is necessary because the kernel silently truncates larger requests.
4. A 64-line `LD_PRELOAD` shim turns off fault read-around on the model mapping, which no stock llama.cpp option does on single-node machines.

The evaluation toolchain records ground-truth routing from llama.cpp, computes exact LRU miss-ratio curves, simulates policies against a Belady optimum, and runs llama.cpp under cgroup memory limits.

On a 15 GB laptop running OLMoE-1B-7B (4.2 GB) under memory limits of 1.5 GB and 2.5 GB, moepager's configuration decoded **1.40 and 3.21 tokens/s**, versus 0.35 and 0.87 tokens/s for the same engine served by the default page cache. That is **3.7–4.0× faster**, with **7–8× less SSD traffic** and memory-pressure stall reduced from 12.7 % to 1.0 %. Default llama.cpp, by contrast, was OOM-killed at both limits unless allowed to spill 2.3–2.5 GB into compressed RAM swap. On a second model, gpt-oss-20b (12.1 GB), replaying real

routing under 4 GB and 6 GB limits showed the same pattern: 1.6–2.0× faster than the kernel default, with 3–4× less data read.

We also report what did not work: a value-based pinning policy captured little of the optimal policy's 34–74 % miss reduction, and completion readahead without turning off read-around made performance worse.

1.2 System architecture

moepager is a **user-space control loop around the Linux page cache**. It sits beside an unmodified inference engine, observes the engine only through the page cache they share, and acts only through standard kernel interfaces. Figure 1 shows the whole system; the subsections below define its structure, components, interfaces, data model, runtime behaviour, process and privilege model, and the design decisions behind them.

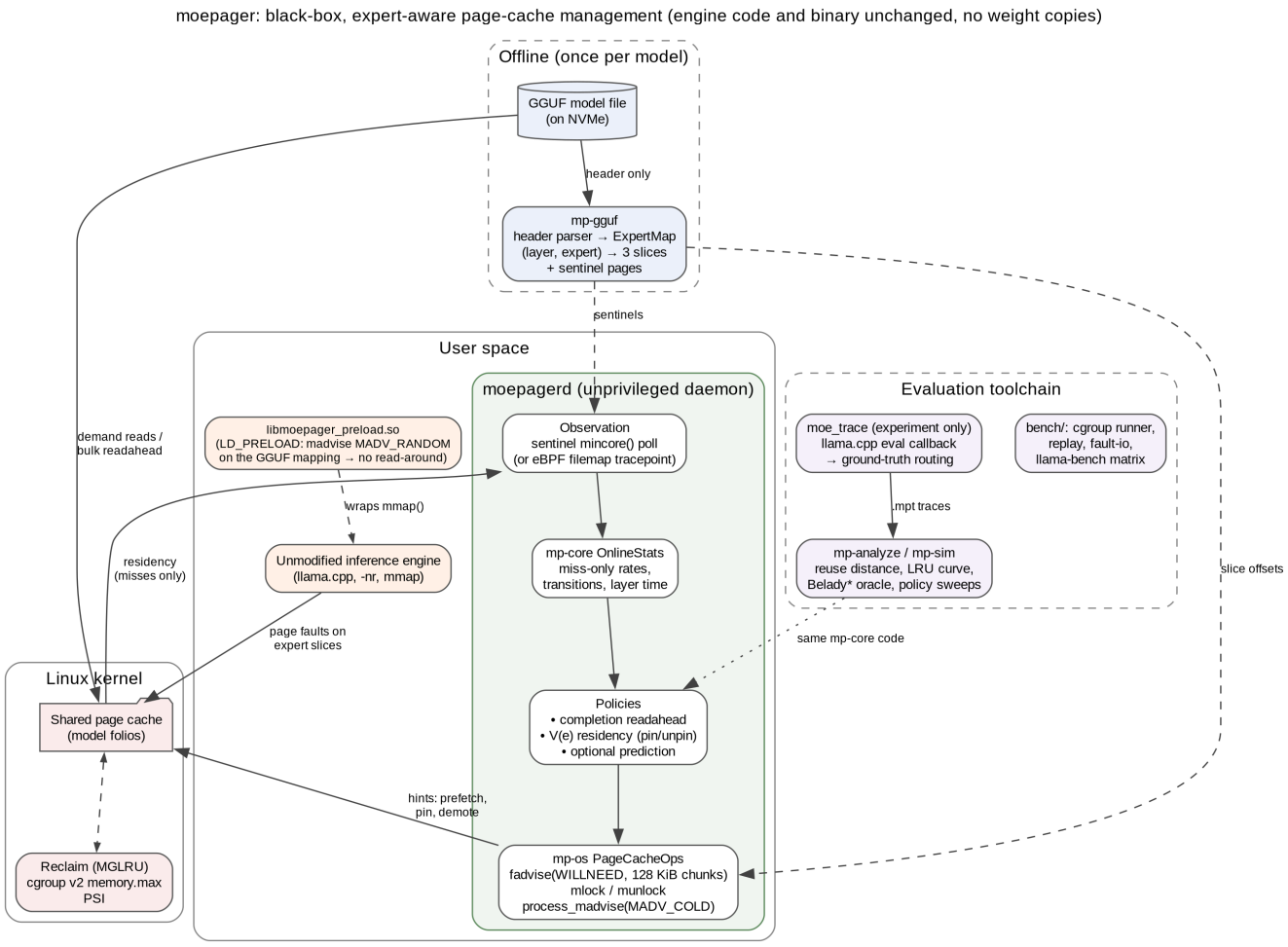


Figure 1. moepager architecture (source: `docs/report/architecture-portrait.dot`; a landscape version is `architecture.png`).

1.2.1 Architectural style

- **Layered, with ports and adapters.** Policy code is pure: it consumes observations and emits `Action` values, and never calls the kernel. All operating-system effects sit behind two traits (`PageCacheOps`, `ResidencyProbe`), with a Linux implementation and a mock one. This lets the same policy run in the simulator, in tests and in the daemon.
- **One core, two hosts.** The policy crate `mp-core` is linked both into the trace-driven simulator (`mp-sim`) and into the live daemon (`moepagerd`), so a policy evaluated offline is exactly the policy deployed (ADR

0001).

- **Three planes.**
  - An **offline plane** runs once per model file: it parses the header and builds the expert map.
  - An **online plane** is the observe → decide → act loop of the daemon.
  - An **evaluation plane** records ground truth, simulates policies against an optimal bound, and benchmarks under memory limits.
- **Non-invasive.** Nothing in the engine's source or binary is changed. The only intervention in the engine's process is an optional `LD_PRELOAD` shim that adds one `madvise` call after `mmap`.

The crate layers, from Cargo dependencies (each layer uses only layers below it):

```
 L4  Front ends     moepager (CLI: gguf-map, synth, analyze, sim, record, replay, fault-io,
…)
                   moepagerd (daemon)
 L3  Evaluation     mp-sim (simulator)   mp-analyze (reuse, MRC)   mp-synth (synthetic
traces)
 L2  Policy core    mp-core: OnlineStats, CostModel, ResidencyEngine, Prefetcher -> Action
 L1  Adapters       mp-gguf (file format)   mp-recorder (observation)   mp-os (kernel calls)
 L0  Data           mp-trace (.mpt trace format, ExpertEvent)
 ------------------------------------------------------------------------------------------
     Linux 6.x: page cache, mincore, posix_fadvise, madvise, mlock, process_madvise,
                cgroup v2, PSI
```

1.2.2 Components and responsibilities

Component	Language / size	Responsibility	Needs privilege
`mp-gguf`	Rust	Parse the GGUF header; build the expert map (units, slices, sentinels, page→unit index); report which tensors llama.cpp would repack	no
`mp-trace`	Rust	`.mpt` binary trace format; CSV import; token inference from layer order	no
`mp-os`	Rust	Kernel interfaces: `mincore`, `cachestat`, chunked `WILLNEED`, `MADV_POPULATE_READ` worker, `mlock`, `process_madvise`; `/proc` parsing (maps, smaps VmFlags, cgroup)	only for pin / demote
`mp-recorder`	Rust	Observation: sentinel polling, full residency scans, bpftrace event conversion, trace replay on a real file	bpftrace only
`mp-core`	Rust	Online statistics (decayed miss rates, layer transitions), cost model, V(e) residency planner, completion and predictive prefetch → `Action`s	no
`mp-sim`	Rust	Byte-capacity cache simulator with an I/O channel; evictors LRU, LFU, Belady*; controllers wrapping `mp-core`	no
`mp-analyze`	Rust	Reuse distances (Fenwick tree), exact LRU miss-ratio curve, locality statistics	no
`moepagerd`	Rust	The daemon: event source → `mp-core` → `PageCacheOps`; start-up checks (read-around, cgroup, memlock limit)	optional
`moepager`	Rust	CLI front end for all offline and evaluation tasks	no
`libmoepager_preload.so`	C, 64 lines	`LD_PRELOAD` shim: `MADV_RANDOM` on read-only shared `*.gguf` mappings	no
`moe_trace`	C++, 169 lines	*Experiment only*: records ground-truth routing from llama.cpp's eval callback	no
`bench/`	Bash	Experiments A–D under cgroup v2 memory walls; metric snapshots	no (user systemd scope)
`python/moepager_tools`	Python	Result collection, report figures	no
llama.cpp [1]	external, unmodified	The inference engine; maps the model with `mmap`	no

1.2.3 Interfaces

The boundaries between layers are four Rust traits and one value type:

```
/// mp-os: actuators on byte ranges of the model file (LinuxOps, MockOps).
pub trait PageCacheOps {
    fn prefetch(&mut self, offset: u64, len: u64) -> io::Result<()>; // async readahead
    fn pin(&mut self, offset: u64, len: u64) -> io::Result<()>;      // resident +
unevictable
    fn unpin(&mut self, offset: u64, len: u64) -> io::Result<()>;
    fn demote(&mut self, offset: u64, len: u64) -> io::Result<()>;   // evict first
}

/// mp-os: residency queries (MappedFile via mincore, MockOps).
pub trait ResidencyProbe {
    fn resident(&self, first: u64, n: u64, out: &mut Vec<bool>) -> io::Result<()>;
    fn page_size(&self) -> u64;
}

/// moepagerd: where observations come from (SentinelSource, TraceSource).
pub trait EventSource {
    fn poll(&mut self) -> io::Result<Option<Vec<ExpertEvent>>>;
}

/// mp-core -> OS layer or simulated cache; units are dense ids l*E + e.
pub enum Action { Pin(u32), Unpin(u32), Prefetch(u32), Demote(u32) }

/// mp-sim: policy hooks driven by the simulator (StaticPin, VControl, ...).
pub trait Controller {
    fn on_access(&mut self, layer: u16, expert: u16, unit: u32, hit: bool, t_ns: u64);
    fn after_layer(&mut self, layer: u16, view: &View) -> Vec<Action>;
    fn end_token(&mut self, view: &View) -> Vec<Action>;
}
```

- The daemon is generic over `PageCacheOps` (`Daemon<O: PageCacheOps>`); `--ops mock` only logs actions, `--ops linux` executes them.
- An `Action` names a *unit*; the daemon expands it into the unit's slice byte ranges using the expert map, so policy code never sees offsets.
- In the simulator, controllers that model black-box observation must drop the `hit = true` calls themselves, which is how the miss-only regime is evaluated.
- External interfaces: the GGUF file format [2]; the kernel system calls and `/proc` files listed under `mp-os` [3]; cgroup v2 control files [4]; the `.mpt` trace format (`docs/TRACE_FORMAT.md`); the expert map as JSON (`map.json`).

1.2.4 Data model

Entity	Fields	Notes
`ExpertMap`	arch, page size, data offset, file size, `n_layers` L, `n_experts` E, top-k, `units`, `dense`, warnings	built once per file; serialisable as JSON
`Unit`	layer l, expert e, bytes s_u, `slices`	dense id u = l·E + e; 1,024 units for OLMoE, 6,144 for Qwen3-30B-A3B
`Slice`	kind (`up`, `gate`, `down`, `gate_up`, `*_bias`), tensor name, absolute offset, length	32-byte aligned, so a page may belong to two units
`DenseRegion`	name, layer, offset, length	attention, norms, embeddings: everything that is not an expert
Sentinel	page index, (l, e), reliable flag	middle page of a weight slice, ≥ 128 KiB from its edges
`PageIndex`	sorted intervals + prefix-max of ends	page → `Owner::Expert{unit, slice}` or `Owner::Dense{region}`
`ExpertEvent`	`t_ns` u64, `token` u32, `layer` u16, `expert` u16	16-byte `.mpt` record; "used" (full) or "missed" (miss-only)
`.mpt` file	magic `MPTRACE\0`, version, record kind, JSON header (model, observation regime, units), fixed-size records	expert records (16 B) or page records (24 B)
`OnlineStats`	decayed counts and exposures per unit, layer-transition rows, layer time	the daemon's only state besides the pinned set

1.2.5 Runtime data flow

For each decoded token, with the recommended configuration (engine with `-nr` and the shim; daemon with `--policy v --budget-mb 0`):

1. The engine computes layer l and reads the k experts the router chose. Pages of the chosen units that are not resident cause page faults. Because of `MADV_RANDOM`, each fault reads only the faulting page.
2. The faulted page is inserted into the shared page cache. If it is a sentinel, the unit's sentinel flips from non-resident to resident.
3. Within one poll period (300 µs in the experiments), `SentinelSource` reports the flip as an `ExpertEvent` for (l, e).
4. `moepagerd` updates `OnlineStats` and infers token boundaries from the layer order.
5. The prefetcher returns `Action::Prefetch(u)`. `LinuxOps` issues `posix_fadvise(WILLNEED)` for every slice of u in 128 KiB chunks, and the kernel starts large asynchronous reads.
6. The engine's remaining faults on u are satisfied by pages that are already resident or already under I/O. They wait on the in-flight read instead of issuing their own small reads.
7. If a pin budget is set, every 4 tokens the residency planner may emit `Pin` / `Unpin` / `Demote` actions, which become `mlock` / `munlock` / `process_madvise(MADV_COLD)`.
8. Under the cgroup's `memory.max`, the kernel's reclaim (MGLRU [5]) evicts unpinned pages as usual; the daemon never evicts by itself unless demotion is enabled.

Offline, `mp-gguf` produces the map used in steps 2–5. For evaluation, `moe_trace` replaces steps 2–3 with exact router output, and `mp-sim` replaces steps 1, 6 and 8 with a model of the cache and the I/O device.

1.2.6 Process, privilege and memory view

Process	Runs	Maps the model	Privilege
Engine (`llama-cli`, `llama-bench`, ...)	unmodified binary, optionally `LD_PRELOAD` shim	`MAP_SHARED`, `PROT_READ`, whole file	none
`moepagerd`	separate process, same user	`MAP_SHARED`, `PROT_READ`, whole file (for `mincore` and `mlock`)	none for observation and prefetch; `CAP_IPC_LOCK` for pinning; `CAP_SYS_NICE` for demotion of the engine's pages

- **No copies.** Both processes map the same file, so they share the same page-cache folios. A page the daemon prefetches or pins is the page the engine reads.
- **One memory budget.** Both run in one cgroup v2 scope with a `memory.max` limit [4]. Page-cache pages are charged to the cgroup of the task that brought them in, so co-location makes the daemon's prefetches count against the same budget as the engine's faults (ADR 0008). The daemon warns when it is not in the engine's cgroup.
- **Start-up checks** (`--pid`): the engine's mapping must carry the `rr` (random-read) VmFlag in `/proc/<pid>/smaps`, otherwise read-around is still on; the memlock limit is compared with the pin budget.
- **Failure isolation.** If the daemon stops, the engine continues with plain kernel behaviour; every action is a hint, and pins are released when the daemon's mapping is closed.

1.2.7 Key design decisions

ADR	Decision	Reason
0001	Policy core in Rust, shared by simulator and daemon	what is simulated is what runs
0002	Treat black-box observation as miss-only	hits are served by the MMU and are invisible to user space
0003	Residency via `mlock` pin + `MADV_COLD` demote, not `fadvise(DONTNEED)`	`DONTNEED` does not drop pages another process maps
0004	Measure (go/no-go, kill criteria) before building the live daemon	avoid optimising a cost that does not exist
0005	128 KiB guard distance for sentinel pages	a neighbour's read-around must not trigger a sentinel
0006	Chunk `WILLNEED` into 128 KiB requests	the kernel silently truncates larger requests
0007	Require `--no-repack` / `-nr` for llama.cpp	repacked weights live in anonymous memory, outside the page cache
0008	Run daemon and engine in one cgroup	fair accounting of prefetched pages
0009	Two-tier observation: unprivileged sentinels, optional bpftrace	works without root; exact events when root is available
0010	Token boundaries from layer regression with a slack	the engine exposes no token signal
0011	Optional `LD_PRELOAD` shim to turn off read-around	no stock llama.cpp option does it on single-node machines

2. Introduction and Problem Statement

2.1 Background

Mixture-of-Experts (MoE) language models [6, 7] replace each dense feed-forward block with many smaller "expert" networks plus a router. For every token, the router picks a few experts per layer (top-k), so only a small fraction of the parameters is used per token. In the three models studied here, OLMoE [8], gpt-oss [9] and Qwen3 [10] (values read from the real model file headers [11]):

Model	File (quantisation)	Size	Layers × experts	top-k	Expert share of file
OLMoE-1B-7B-0924	Q4_K_M	4.21 GB	16 × 64	8	92.6 %
gpt-oss-20b	MXFP4	12.11 GB	24 × 32	4	84.1 %
Qwen3-30B-A3B	Q4_K_M	18.56 GB	48 × 128	8	94.6 %

Because only 4–8 experts per layer are touched per token, MoE models are an attractive fit for consumer machines. A common way to run a model larger than free RAM is llama.cpp [1] in `mmap` mode: the model file is mapped into memory, and the Linux page cache loads weights on demand and evicts them under memory pressure.

2.2 Problem statement

The kernel's page cache is generic. Its reclaim policy (MGLRU [5]) and its fault read-around [12] know nothing about how a GGUF file [2] is laid out or how MoE routing behaves. Our measurements on a 15 GB laptop (Section 6) show three concrete problems:

1. **The unit of use is scattered.** One expert at one layer is stored as three slices in three different tensors (`ffn_up_exps`, `ffn_gate_exps`, `ffn_down_exps`), hundreds of megabytes apart. The kernel cannot know these three ranges belong together.
2. **Read-around pollutes the cache.** On a page fault, Linux reads a window around the faulting page: 4 MiB on this machine's btrfs filesystem. For expert slices of about 1–1.3 MB, most of that window belongs to neighbouring experts that will not be used. Replaying real OLMoE routing under a 1.5 GB budget, the SSD read **1.80 GB per token to deliver 0.49 GB of needed expert weights** (3.7× amplification), and almost every expert access missed.
3. **Fault-driven reads are slow.** Faulting cold experts in through mmap delivered only 0.15–0.24 GB/s on the real model file. Reading the same bytes with large prefetch requests delivered 1.5–1.7 GB/s.

A further, practical problem: llama.cpp on x86 CPUs with AVX2 by default *repacks* the most common quantised expert weights into ordinary (anonymous) memory. That is 77 % of the expert bytes for OLMoE and Qwen3 Q4_K_M, and 100 % for gpt-oss MXFP4. Under memory pressure those pages go to swap instead of the page cache.

Goal: make MoE inference faster on memory-constrained Linux machines, using knowledge of the model file's structure and of routing behaviour, **without modifying the inference engine** and without copying the weights.

3. Literature Review and Objectives

3.1 Related work

MoE offloading inside the inference engine. Most systems that keep only part of an MoE model in fast memory are built into the engine, where the router's decisions are directly visible:

- mixtral-offloading (Eliseev and Mazur [13]): LRU expert cache plus speculative prefetch.
- MoE-Infinity (Xue et al. [14]): activation-aware expert cache.
- HOBBIT (Tang et al. [15]): mixed-precision experts, implemented as about 8,000 lines added to llama.cpp.
- Pre-gated MoE [16], DuoServe-MoE [17] and similar systems: predict next-layer experts and prefetch them. Liu et al. [18] survey this design space.

Inside llama.cpp itself there are several open proposals [19]: PR #25294 (O_DIRECT expert streaming that bypasses the page cache), PR #26824 (expert heat map and mmap pinning), PR #29887, issue #20757 and discussion #27149. All of these modify the engine.

Using the OS page cache as the expert tier. Si et al. [20] is the closest study. On a datacenter GH200 node they show that kernel recency (LRU) serves about as much demand as an oracle static-frequency pin, that Belady's bound [21] leaves about one third of LRU's misses addressable, and that `fadvise(WILLNEED)` prefetch hints gained only 0.3–5 %. Their engine reads with `pread`, so mmap fault behaviour and file structure play no role.

Operating-system mechanisms.

- cache_ext / cachebpf (Zussman et al. [22]) lets eBPF programs replace page-cache eviction, but needs a patched kernel.
- MGLRU [5] and DAMON [23] are generic and know nothing of model structure.
- vmtouch [24] pins or evicts file ranges given by the user.
- "LLM in a flash" (Alizadeh et al. [25]) bundles co-used weights so that flash reads are large. That is the same idea as our expert-completion readahead, but it is implemented inside the engine and requires reformatting the model. PowerInfer [26] similarly exploits activation locality inside the engine, keeping frequently activated ("hot") neurons in fast memory.

Gap. We found no system that manages the page cache for an *unmodified* engine using the *model file's structure*, and none that measures the effect of mmap fault read-around on MoE experts.

3.2 Objectives

- **O1.** Build an exact map from GGUF files to (layer, expert) units and their byte and page ranges, verified on real models.
- **O2.** Observe expert usage from outside the engine, without root.
- **O3.** Build an evaluation toolchain: ground-truth routing traces, trace analysis, a policy simulator with an optimal (Belady) bound, real-kernel replay, and benchmarks under a fixed memory budget.
- **O4.** Implement page-cache actuators: chunked readahead, pinning and demotion. Combine them in a daemon whose policy code is identical to the simulated one.
- **O5.** Decide go/no-go on real hardware with predefined kill criteria, and report only measured numbers.

4. Methodology and Algorithms

This section defines the problem formally (4.1), then describes each algorithm in the order data flows through the system: the expert map (4.2), black-box observation (4.3), read-around suppression (4.4), expert-completion readahead (4.5), the optional residency policy (4.6) and the offline evaluation algorithms (4.7). Sections 4.8–4.10 give the experimental protocol, the metric definitions and the threats to validity. Every algorithm below corresponds to code in the repository; the crate is named in each heading.

4.1 Problem model and notation

Symbol	Meaning
F, P	the GGUF model file; the page size (4 KiB on the test machine)
L, E, k	number of MoE layers, experts per layer, experts chosen per token per layer (top-k)
u = (l, e)	an expert **unit**: expert e of layer l; U is the set of all L·E units
S(u)	the slices of u: one byte range `[a, b)` of F per expert tensor (`up`, `gate`, `down`, optional bias rows)
s_u	size of unit u in bytes, Σ over S(u) of (b − a)
R_{t,l}	the k experts the router selects at decode step (token) t, layer l
σ	an access trace: the sequence of units R_{1,1}, R_{1,2}, …, R_{T,L}
C	the memory budget: cgroup v2 `memory.max` [4]
B_d, B_b	delivered bandwidth of fault-driven reads and of bulk readahead
t_f	fixed cost of one demand miss (fault entry plus first I/O latency)

Decoding token t touches, layer by layer, every unit in R_{t,l} plus the dense (non-expert) tensors of layer l. The engine maps F read-only and shared, so every touched byte is served from the page cache, which holds at most C bytes (minus the engine's anonymous memory). A unit access is a **hit** if all its pages are resident and a **miss** otherwise.

The quantity to minimise is decode time per token. Under memory pressure it is dominated by I/O stall, which a simple cost model captures:

```
stall(miss of u)  =  t_f + s_u / B_d              (kernel default)
stall(miss of u)  =  t_f + t_detect + s_u / B_b     (with completion readahead)
```

Two levers follow directly. **Residency** (which units stay cached) reduces the number of misses. **Read path** (how a missing unit is read) reduces the cost of each one. The original hypothesis was that residency is the main lever; Section 6 shows that on an `mmap` engine the read path dominates, because B_d is small and the kernel's read-around inflates the bytes read per miss.

4.2 Expert map from the GGUF header (`mp-gguf`)

A GGUF file [2] starts with a header listing every tensor's name, shape, ggml type and data offset. `mp-gguf` parses only this header, so a truncated prefix of the file is enough. For each tensor named `blk.<l>.ffn_{up,gate,down,gate_up}_exps.weight` with shape `[ne0, ne1, E]`, the expert index is the outermost (slowest-varying) dimension, so each expert's weights are contiguous inside the tensor:

```
Algorithm 1  BuildExpertMap(header)
  for each tensor T matching blk.<l>.ffn_*_exps.*:
      bytes  := type_size(T.type) * (ne0 * ne1 * E) / block_size(T.type)
      stride := bytes / E                       # warn if E does not divide bytes
      for e in 0 .. E-1:
          add slice [T.offset + e*stride, T.offset + (e+1)*stride) to unit (l, e)
  for each 2-D per-expert tensor ([ne0, E], e.g. *_exps.bias):
      add one row per expert to unit (l, e)
  sort all slices by offset; build page->unit interval index (below)
  for each slice: choose a sentinel page (Section 4.3)
```

- **Page → unit index.** Slices are 32-byte aligned rather than page aligned, so a boundary page can belong to two units. The index stores slices as intervals sorted by start, plus a prefix-maximum array of interval ends, so the owners of any page are found by binary search in O(log n + r), where r ≤ 2 is the number of owners.
- **Repacking check.** `mp-gguf` also reports which expert tensors llama.cpp's CPU backend would repack into anonymous memory, based on the quantisation type and the repack rules in the engine source [1]. On AVX2 this covers 77 % (Q4_K_M) to 100 % (MXFP4) of the expert bytes, so the engine must run with `--repack 0` / `-nr` for the weights to stay in the page cache.
- **Cost.** O(number of tensors) to parse, O(n log n) to sort the n slices (n = 3·L·E; 3,072 for OLMoE, 18,432 for Qwen3-30B-A3B).
- **Verification.** Fixture GGUF files with known layouts are checked slice by slice (kind, offset, length of every unit), the page index is checked against the slices, and the real headers of OLMoE-1B-7B [8], gpt-oss-20b [9] and Qwen3-30B-A3B [10] are checked for layer, expert and top-k counts and unit sizes.

4.3 Black-box observation by sentinel pages (`mp-recorder`)

The engine is a black box: the daemon cannot see router decisions. It infers them from the page cache, which is shared.

Sentinel choice. For each slice, one page in its middle is chosen as the **sentinel**, at least 128 KiB (`SENTINEL_GUARD`) from either edge of the slice. A fault in a *neighbouring* slice brings in a read-around window around the faulting page; with the default 128 KiB window, that window cannot reach this sentinel. Slices shorter than 2 × 128 KiB + 2P are marked unreliable, and bias rows (a few KiB, inside other tensors' pages) get no sentinel. On btrfs the window is 4 MiB, larger than a whole OLMoE slice, which is why read-around must be turned off (Section 4.4) for observation to be correct as well as fast.

Polling. The daemon maps F read-only (`MAP_SHARED`, `PROT_READ`) and calls `mincore(2)` [3] on the sentinel pages. This needs no privilege and works for files the daemon does not own. (`cachestat(2)` would need write access to the file.)

```
Algorithm 2  SentinelTick(sentinels, prev, now_ns)       # every Δ = 300 µs
  events := []
  for i, s in enumerate(sentinels):
      r := mincore(s.page)
      if r and not prev[i] and (s.layer, s.expert) not in events:
          events.append((now_ns, s.layer, s.expert))      # a miss of unit (l, e)
      prev[i] := r
  return events
```

- **Semantics.** A sentinel changing from non-resident to resident means "unit (l, e) has just been faulted in", which is a **miss**. Hits are invisible: pages already mapped are read through the CPU's page tables

without entering the kernel, so no user-space interface reports them. All online statistics are therefore *miss-only* (ADR 0002).
- **Cost.** One `mincore` call per sentinel per sweep: O(3·L·E) system calls every Δ. The poll period Δ bounds detection latency to at most Δ plus the remaining sweep time.
- **False positives.** A sentinel can also turn resident because of the daemon's own prefetch of the same unit, or because read-around from an unrelated fault reached it. The first case is harmless (it re-issues a prefetch for a unit being read anyway). The second is eliminated by turning read-around off.
- **Privileged alternative.** With root, the same events can be taken exactly from the `filemap:mm_filemap_add_to_page_cache` tracepoint; a bpftrace script is included (ADR 0009). It was not used for the results.

Token boundaries are not visible either. Within one decode step layers are visited in increasing order, so a new token is inferred when an event's layer falls more than a slack (default 2 layers) below the highest layer seen in the current step:

```
Algorithm 3  InferToken(event)                              # ADR 0010
  if max_layer is set and event.layer + slack < max_layer:
      end_token()                       # closes statistics, may trigger re-plan
  max_layer := max(max_layer, event.layer)
```

The slack absorbs reordering when several layers are discovered in the same sweep. On ground-truth traces this recovers the true token count; on miss-only traces a token whose misses are all in high layers can merge with the previous one, which only affects the V(e) rate estimates.

4.4 Read-around suppression (`tools/preload`)

On a page fault in a file mapping, Linux reads a window of pages around the faulting address ("read-around"), sized from the backing device's `read_ahead_kb` (4 MiB for this machine's btrfs). For a 1–1.3 MB expert slice, most of that window is neighbouring experts that will not be used. The waste has two effects: up to 3.7–4.8× more SSD traffic than needed, and a cache filled with unused pages, which raises the miss ratio.

Setting `MADV_RANDOM` on the mapping (`VM_RAND_READ` in the kernel) disables read-around for it, so a fault reads exactly the faulting page. Stock llama.cpp applies this only when `--numa` is set on a multi-node machine. A 64-line `LD_PRELOAD` shim wraps `mmap()` and, for read-only shared mappings of `*.gguf` files, calls `madvise(addr, len, MADV_RANDOM)` after the original `mmap` returns (ADR 0011). The engine's code and binary are unchanged. The daemon verifies the effect by checking for the `rr` flag in the engine's `/proc/<pid>/smaps` VmFlags and warns if it is missing.

Turning read-around off alone trades one problem for another: each miss becomes hundreds of synchronous 4 KiB faults (39,000 per token in Experiment B). Completion readahead (next section) turns them back into bulk reads, but only for the unit that was actually chosen.

4.5 Expert-completion readahead (`moepagerd`, `mp-os`)

When a miss of unit u is observed, the daemon immediately asks the kernel to read **all** of u asynchronously. The engine is typically still inside the first slice (`up`); the `gate` and `down` slices, hundreds of megabytes away in the file, arrive as large reads instead of thousands of small faults.

```
Algorithm 4  OnMiss(u)                                        # mp-core::prefetch
  for each slice [a, b) in S(u):
      for c in a, a+128K, a+256K, ... < b:                     # chunked (ADR 0006)
          posix_fadvise(fd, c, min(128K, b - c), POSIX_FADV_WILLNEED)
  stats.observe(u, MISS)
```

- **Chunking.** A single `WILLNEED` is silently truncated by `force_page_cache_ra()` in `mm/readahead.c` [12] to max(`io_pages`, `ra_pages`) pages, so a 1 MiB request populated only 256 KiB in our tests. Issuing 128 KiB chunks makes every byte requested actually read.
- **Asynchrony.** `WILLNEED` queues reads and returns, so the daemon never blocks on I/O. The alternative `MADV_POPULATE_READ` (`--prefetch-mode populate`) is synchronous and runs on a worker thread; it is for devices where readahead is disabled and `WILLNEED` has no effect.
- **Prediction-free.** The action is taken only for units the router has already chosen, so it never wastes bandwidth on wrong guesses. A cross-layer predictive variant (`--predict`) exists but is not recommended (Section 6).
- **Repeated misses.** Each unit has three sentinels, so one real miss can trigger up to three prefetches of the unit. A refractory window that suppressed repeats within 20 ms was evaluated and *doubled* misses (Phase 7 ablation: decode speed fell from 5.4 to 2.1 tok/s), because the repeats re-fetch pages reclaimed while the engine is still working through the unit. The window is therefore off by default.
- **Cost.** O(s_u / 128 KiB) system calls per miss, about 30 for an OLMoE unit; no data is copied, and the pages are shared with the engine.
- **Memory accounting.** Pages read on the daemon's behalf are charged to the daemon's cgroup. Daemon and engine are therefore run in the same cgroup (ADR 0008), so prefetching cannot exceed the budget C.

4.6 Residency policy V(e) (`mp-core`; optional, needs privilege)

With `CAP_IPC_LOCK` (or a raised `RLIMIT_MEMLOCK`), the daemon can pin units with `mlock` through its own shared mapping, which pins the same page-cache folios the engine reads. The policy chooses a pinned set within a pin budget C_pin ≤ C by **value per byte**:

```
    V(u) = rate(u) · (t_f + s_u / B_d) / s_u
```

that is, the expected stall avoided per token per byte held. This is a fractional-knapsack relaxation; the greedy order is optimal for the relaxation and near-optimal when units are small compared with C_pin.

Rate estimator under miss-only observation. A pinned unit produces no misses, so naive miss counts would make every pinned unit look cold. rate(u) is instead defined as the **miss rate per token while unpinned**:

```
    rate(u) = (Σ_t w_t · miss_t(u) + p·k) / (Σ_{t: u unpinned at t} w_t + k)
```

- w_t = g^t with g = 2^{1/h} is an exponential decay weight with half-life h = 64 tokens, implemented by growing the weight of new events instead of shrinking old ones (O(1) per event, renormalised when w exceeds 10^100).
- p is a prior rate and k = 0.05·w_t a light shrinkage, so a unit exposed for one token is not estimated at exactly 0 or 1.
- While a unit is pinned its estimate is frozen. At each re-plan a random 2 % of pinned units are released ("exploration") so frozen estimates are refreshed.

```
Algorithm 5  Replan(stats, cost)                  # every 4 tokens (replan_every)
  excluded := random 2 % of pinned units          # exploration, miss-only only
  cand := [(V(u) * (1.25 if pinned[u] else 1), u) for u not excluded, V(u) > 0]
  sort cand by value descending                   # hysteresis factor 1.25
  want := {}; used := 0
  for (v, u) in cand:
      if used + s_u <= C_pin: want += u; used += s_u
  emit Unpin(u) [+ Demote(u)] for u pinned and not in want     # unpins first,
  emit Pin(u) for u in want and not pinned                     # so C_pin holds
```

- **Cost.** O(n log n) per re-plan for n = L·E units (1,024 for OLMoE), amortised over 4 tokens.
- **Hysteresis.** A pinned unit's value is multiplied by 1.25 when ranking, so a newcomer must beat it by 25 % to displace it. This bounds pin/unpin churn.
- **Demotion.** Units that leave the pinned set can be marked cold with `process_madvise(pidfd, MADV_COLD)` on the engine's mapping (needs `CAP_SYS_NICE`). `posix_fadvise(DONTNEED)` was rejected because it does not drop pages that another process has mapped (verified on Linux 6.19; ADR 0003).

The policy code is one Rust implementation used by both the simulator and the daemon (ADR 0001), so simulated and live behaviour cannot drift apart.

4.7 Offline evaluation algorithms (`mp-sim`, `mp-analyze`)

Trace-driven simulation. `mp-sim` replays a trace σ against a cache of C bytes:

- a hit costs nothing;
- a miss is queued on a single FIFO I/O channel and costs `t_f + s/B_d`, or `t_f + t_detect + s/B_b` with completion readahead;
- compute time per layer is taken from the trace timestamps;
- the reported quantities are miss ratio, bytes read, and modelled stall.

Policies compared: LRU (a stand-in for the kernel's MGLRU [5]), LFU, prefix pinning, an oracle static-frequency pin, V(e) with full or miss-only observation, predictive prefetch, and **Belady***.

Belady* evicts the cached unit whose next use is furthest in the future [21]. For equal sizes this is optimal (MIN). With variable sizes exact optimality is NP-hard, so the same rule is used as the standard heuristic and labelled with an asterisk. Implementation: for each unit, the sorted list of its positions in σ; the next use after position i is found by binary search; cached units are kept in an ordered set keyed by next use, so each access costs O(log N).

Exact LRU miss-ratio curve. `mp-analyze` computes, independently of the simulator, the miss ratio of a byte-capacity LRU cache at *every* capacity in one pass, using stack-distance analysis [27] with a Fenwick tree [28]:

```
Algorithm 6  ByteReuseDistances(σ)
  fenwick := zeros(N)                      # one slot per trace position
  for i, u in enumerate(σ):
      if last[u] exists:
          D_i := fenwick.sum(last[u]+1 .. i-1)   # bytes of distinct units since last use
          fenwick.add(last[u], -s_u)
      else:
          D_i := infinity                        # cold miss
      fenwick.add(i, +s_u); last[u] := i
  # access i hits in LRU of capacity C  <=>  D_i + s_u <= C
  MRC(C) := |{ i : D_i + s_u > C }| / N         # sort thresholds once, O(N log N)
```

Each position holds the size of the unit whose *latest* access is there, so a range sum counts each distinct unit once. The whole curve costs O(N log N). The simulator's LRU agrees with this curve exactly at every tested capacity; this cross-check between two independent implementations is a regression test.

4.8 Experimental protocol

Hypotheses and kill criteria. The project was run as a go/no-go experiment (ADR 0004) with criteria fixed before measuring (IDEA_REVIEW §6):

- **K1:** Belady* cannot save at least 15 % of misses relative to LRU at any realistic capacity → residency management is pointless.
- **K2:** fault-driven bandwidth is at least 0.75× bulk bandwidth → the read path is not worth changing.
- **K3:** default llama.cpp beats every page-cache configuration at equal memory → the approach has no user.

Experiments.

	Question	Independent variables	Measured
A	How local and predictable is real routing; how much could any policy gain?	prompt domain (chat, code, prose); cache size 15–70 % of expert bytes; policy	reuse, concentration, next-layer recall; miss ratio per policy
B	What does the real kernel do with this access pattern?	memory budget (1.5, 2.5 GB); read-around on/off; daemon on/off	tokens/s, SSD bytes/token, major faults/token
C	How fast are fault-driven vs bulk reads of expert units?	read method (fault, WILLNEED one call, WILLNEED 128 KiB chunks, pread 1 MiB)	GB/s delivered, read amplification
D	Does it help a real engine end to end?	budget; configuration B0, B0S, B2, D2, P, PD	decode tokens/s, SSD bytes/token, faults/token, memory PSI

- **A: ground truth.** `moe_trace`, a 170-line experiment-only program, records the router's choices through llama.cpp's scheduler eval callback by reading the `ffn_moe_topk-<l>` tensors row by row. Three prompts × 255 decode tokens give 32,640 unit uses per prompt for OLMoE (255 × 16 layers × top-8). The traces are then analysed (4.7) and simulated at seven capacities.
- **B: real-kernel replay.** `moepager replay` touches, for each token of the prose trace, every page of every chosen unit through `mmap` with 4 threads (64 KiB chunks interleaved). It isolates the kernel's paging behaviour from the engine's compute.
- **C: bandwidth.** 32 cold, randomly chosen expert-sized ranges of the real file per method, 3 seeds, 8 threads.
- **D: end to end.** `llama-bench -p 0 -n 32 -r 3 -t 8 -lm mmap` [1]. Configurations: B0 default (repacking, zram swap allowed); B0S default without swap; B2 `--repack 0`; D2 B2 + moepagerd; P B2 + read-around off (shim); PD P + moepagerd.

Controls applied to every run:

- **Cold start:** the model's pages are dropped with `posix_fadvise(DONTNEED)` while no process maps it, followed by `sync` and a 5 s settle.
- **Memory wall:** each run executes in a transient cgroup v2 scope (`systemd-run --user --scope -p MemoryMax=C -p MemorySwapMax=0`) [4]. Swap is allowed only for B0 (`MemorySwapMax=infinity`). Daemon and engine share the scope.
- **Same binary, same file:** one llama.cpp build (commit 970b51c), one GGUF file per model, CPU only, 8 threads (`nproc` − 4).
- **Environment capture:** kernel version, CPU, free memory, filesystem type, every bdi's `read_ahead_kb`, MGLRU state and the llama.cpp commit are saved with each result set (`env.txt`).

- **Isolation of runs:** no test suite or build ran concurrently with measurements; one set of D runs that overlapped a test run was discarded and repeated.

4.9 Metric definitions

Metric	Definition	Source
tokens/s (D)	llama-bench `tg` mean ± standard deviation over repetitions	llama-bench output
tokens/s (B)	replayed tokens ÷ wall time	replay tool
SSD bytes / token (D)	(sectors read on the model's block device after − before) × 512 ÷ tokens; tokens = n_gen × reps + 1 (warm-up)	`/proc/diskstats`
Data read / token (B)	(`pgsteal` × 4 KiB + `file` at exit) ÷ tokens: page-cache inflow charged to the run's cgroup, including the daemon's reads	cgroup `memory.stat`
Major faults / token	D: `pgmajfault` of the cgroup ÷ tokens; B: the replayer's `ru_majflt` ÷ tokens	cgroup `memory.stat`, `getrusage`
Memory PSI	`some avg10` of the cgroup's `memory.pressure` at exit: share of the last 10 s in which at least one task stalled on memory [29]	cgroup
Read amplification	bytes read from storage ÷ expert bytes needed	C: `/proc/self/io` (single process, no daemon)
Miss ratio	misses ÷ unit accesses	simulator, MRC
Belady* saving	1 − misses(Belady*) ÷ misses(LRU) at equal capacity	simulator
Next-layer recall	share of R_{t,l+1} contained in a size-k prediction made after layer l	`mp-analyze`

Only measured numbers are reported. Where a configuration did not finish (OOM kill, timeout), it is reported as such, never as zero or extrapolated.

4.10 Threats to validity

- **Internal.**
  - A desktop session ran during the measurements; background activity adds noise, visible as run-to-run spread in some D configurations.
  - B uses one replay per variant; D uses 3 repetitions of 32 tokens. Differences below about 10 % should not be read as significant.
  - `/home` is btrfs with zstd compression, so device bytes can be lower than logical bytes. Quantised weights compress poorly, and the comparisons are between configurations on the same file.
  - Per-process I/O counters (`/proc/self/io`) do not include reads issued by another process. B's traffic is therefore taken from the cgroup (4.9); an earlier draft used the replayer's own counter and understated the daemon variants by 0.1–0.6 GB/token.
  - B0's swap device is zram, which is RAM. Its "fast" result uses memory outside the 1.5 / 2.5 GB budget, so it is reported separately.
- **Construct.**
  - The replay tool (B) touches pages but does not compute; it measures paging, not inference. D measures the real engine.
  - LRU in the simulator is an approximation of MGLRU [5]; B shows the kernel's behaviour directly.
  - Belady* with variable sizes is a heuristic bound, not a proven optimum.
- **External.**
  - One machine, one SSD, one filesystem. Read-around on ext4 defaults to 128 KiB, so the pollution effect, and the benefit of turning it off, will be smaller there.
  - End-to-end results exist for OLMoE only. gpt-oss-20b has A–C results; its D run was stopped because the B0 configuration exhausted system memory. Qwen3-30B-A3B was mapped from its

header only.
    - Only llama.cpp's CPU backend was studied.
- **Conclusion.** The go/no-go decision depends on K1–K3 as defined before measuring; the unanticipated read-around finding is reported as such rather than folded into the original hypothesis.

5. Implementation

5.1 Technology stack and structure

Layer	Choice	Why
Core, simulator, daemon, CLI	**Rust** (stable), 10 crates, ≈ 8,900 lines	Predictable latency for a daemon; fast enough for simulation sweeps; one language, so the daemon and the simulator share the *same* policy code
OS interface	`libc` syscalls behind traits (`PageCacheOps`, `ResidencyProbe`)	Real Linux implementation + `MockOps` for deterministic tests
Ground-truth capture	C++ against `libllama` (`tools/moe_trace`, 169 lines)	Needs the engine's eval callback; experiment-only
Read-around shim	C, `LD_PRELOAD` (`tools/preload`, 64 lines)	Disables fault read-around on the model mapping without touching the engine's code or binary
Privileged tracing	bpftrace script (26 lines)	Optional, root only
Fixtures, metric parsing, probes	Python (stdlib + pytest), 674 lines	Independent GGUF writer so parser tests aren't self-confirming
Experiments	Bash + cgroup v2 (`systemd-run --user`), no root	Fixed memory budget per run
CI	GitHub Actions: fmt, clippy `-D warnings`, tests, ruff, pytest, demo smoke run	

Crate	Responsibility
`mp-gguf`	GGUF v2/v3 header parser, ggml type table, `ExpertMap`, page index, sentinels, repack-risk detection
`mp-trace`	`.mpt` binary trace format (spec in `docs/TRACE_FORMAT.md`), CSV import/export, token inference
`mp-core`	Deterministic RNG, cost model, `OnlineStats` (miss-only), `ResidencyEngine` (V(e)), `PrefetchPlanner`
`mp-synth`	Synthetic routing generator (skew, reuse, cross-layer affinity, drift)
`mp-analyze`	Fenwick-tree reuse distance, exact LRU miss-ratio curve, routing predictability, timing
`mp-sim`	Trace-driven cache + I/O-channel simulator, policies, Belady*
`mp-os`	`mincore`, `cachestat`, chunked `fadvise`, `mlock`, `process_madvise`, `/proc/pid/maps`/`smaps` parsing
`mp-recorder`	Sentinel and full-scan recorders, page→expert conversion, bpftrace ingest, mmap/pread trace replayer
`moepager`	CLI: `gguf-map`, `synth`, `analyze`, `sim`, `import-csv`, `trace-csv`, `record`, `ingest-bpftrace`, `page2expert`, `replay`, `fault-io`
`moepagerd`	Daemon: policies `none`, `willneed-all`, `lru-pin`, `v`; sources `sentinel` / `trace:`; ops `linux` / `mock`

5.2 Functional completeness

Feature	Status
GGUF expert map, verified on 3 real models	done, tested
Ground-truth routing capture + import	done, used on OLMoE
Trace analysis, simulator, Belady bound	done, tested, used
Unprivileged sentinel observation	done, tested end to end on real files
Chunked completion readahead	done, measured (Section 6)
Read-around shim	done, measured (Section 6)
V(e) pinning	implemented and unit-tested; live use needs `CAP_IPC_LOCK` (default memlock limit is 8 MB), **not measured**
`MADV_COLD` demotion	implemented; needs `CAP_SYS_NICE`, **not measured**
eBPF observation	script + ingest written; needs root, **not run**
Start-up checks: read-around still on, daemon outside the engine's cgroup	done, tested live
`MADV_POPULATE_READ` prefetch fallback (`--prefetch-mode populate`)	done, tested on a real file; not benchmarked
PSI-driven budget control, dashboard	future work

5.3 Correctness

- **Tests:** 87 Rust and 13 Python tests, run in CI. They include:
  - parser checks against GGUF files written by an *independent* Python writer;
  - an assertion that simulated LRU equals the analytically computed LRU curve exactly;
  - a check that Belady* never loses to any other policy;
  - real-kernel tests of `mincore`/`cachestat`/`fadvise`/`mlock` semantics on a real file;
  - end-to-end recorder tests that recover a replayed trace with 100 % recall;
  - daemon dry runs.
- **Real-model checks:**
  - The map of OLMoE/Qwen3/gpt-oss matches tensor sizes in the headers.
  - Each captured (token, layer) has exactly top-k experts.
  - The no-read-around replay reproduced the simulator's LRU prediction (Section 6.3).
- **Bugs found by testing and fixed:**
  - WILLNEED truncation (found by CI);
  - the btrfs device-number mismatch between `stat()` and `/proc/pid/maps`;
  - read-around/compression spill causing false expert detections;
  - duplicate completion prefetches (refractory window).

5.4 Demonstration

`make demo` runs the full pipeline with no model or root. It parses the real Qwen3 map, generates routing, analyzes it, simulates policies, does a daemon dry run, and records a replayed trace black-box. With a model, `bench/phase7.sh` runs experiments A–D. Terminal screenshots of the tools on the real OLMoE model follow.

5.5 Screenshots (real output on OLMoE-1B-7B)

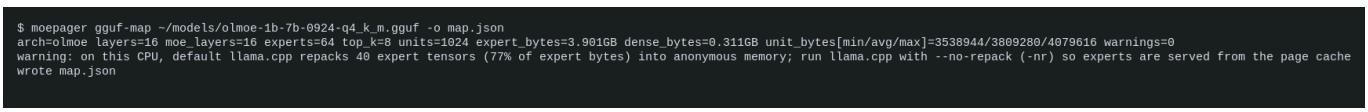


Figure 2. Expert map from the real GGUF header, with the repack warning for this AVX2 CPU.

```
$ ./build/moe_trace/moe_trace -m olmoe-1b-7b-0924-q4_k_m.gguf -o trace-prose.csv -n 256 --no-repack -p "..."
moe_trace: 256 tokens generated, 32640 expert rows -> bench/results/phase7-olmoe/trace-prose.csv
token,layer,expert,t_ns
0,0,6,1106548300
0,0,15,1106548300
0,0,63,1106548300
...
```

Figure 3. Ground-truth routing capture through llama.cpp's eval callback.

```
$ moepager import-csv trace-prose.csv --map map.json -o trace-prose.mpt
imported 32640 rows, 255 tokens -> trace-prose.mpt

$ moepager analyze trace-prose.mpt --map map.json
source=llama-evalcb observation=Full tokens=255 accesses=32640
footprint=2.866GB cold_misses=750 median_reuse_units=Some(193)
token_reuse=0.448 top10_share=0.376 norm_entropy=0.832 units_touched=0.732
next-layer recall@k: transitions=0.687 popularity=0.453 prev-token=0.455 (concentration top4=0.399)
layer gap p10/p50/p90 = 1.507/1.936/3.224 ms; token p50 = 35.624 ms
LRU miss-ratio curve (capacity as % of footprint):
    10%  cap=0.287GB  miss_ratio=1.000  miss_bytes/token=0.488GB
    20%  cap=0.573GB  miss_ratio=0.554  miss_bytes/token=0.271GB
    30%  cap=0.860GB  miss_ratio=0.454  miss_bytes/token=0.223GB
    40%  cap=1.146GB  miss_ratio=0.321  miss_bytes/token=0.157GB
    50%  cap=1.433GB  miss_ratio=0.286  miss_bytes/token=0.140GB
    60%  cap=1.719GB  miss_ratio=0.260  miss_bytes/token=0.127GB
    70%  cap=2.006GB  miss_ratio=0.212  miss_bytes/token=0.104GB
    80%  cap=2.292GB  miss_ratio=0.139  miss_bytes/token=0.068GB
```

Figure 4. Importing the routing trace and analyzing it: reuse, predictability, exact LRU curve.

```
$ moepager sim trace-prose.mpt --map map.json --caps 0.25,0.5 --completion off --policies lru,lfu,static-freq,v-full,v-miss,belady
|trace=bench/results/phase7-olmoe/trace-prose.mpt tokens=255 (warmup 16) expert_bytes=3.901GB t_layer=1.936ms demand=0.45GB/s bulk=2.4GB/s
| capacity | completion | policy | miss ratio | SSD GB/token | est tok/s
|---|---|---|---|---|---
| 25% (0.98 GB) | off | lru | 0.434 | 0.212 | 1.96
| 25% (0.98 GB) | off | lfu | 0.331 | 0.162 | 2.51
| 25% (0.98 GB) | off | static-freq* | 0.375 | 0.185 | 2.23
| 25% (0.98 GB) | off | v-full | 0.375 | 0.184 | 2.24
| 25% (0.98 GB) | off | v-miss | 0.391 | 0.193 | 2.15
| 25% (0.98 GB) | off | belady* | 0.237 | 0.116 | 3.40
| 50% (1.95 GB) | off | lru | 0.222 | 0.108 | 3.61
| 50% (1.95 GB) | off | lfu | 0.080 | 0.039 | 8.27
| 50% (1.95 GB) | off | static-freq* | 0.149 | 0.073 | 5.06
| 50% (1.95 GB) | off | v-full | 0.146 | 0.071 | 5.16
| 50% (1.95 GB) | off | v-miss | 0.151 | 0.074 | 5.01
| 50% (1.95 GB) | off | belady* | 0.058 | 0.028 | 10.32
```

Figure 5. Policy simulation, including the Belady bound.*

```
$ python3 python/moepager_tools/collect.py bench/results/phase7-olmoe   (final pass, 3 reps)
| budget | config | tok/s | SSD GB/token | majflt/token | refault/token | PSI some avg10 | PSI stall ms | peak GB |
|---|---|---|---|---|---|---|---|---|
| D-1500M | B0 | 3.17 | 0.310 | 38205.5 | 64181.4 | 16.21 | 6351 | 1.57 |
| D-1500M | B0S | TBD | TBD | TBD | TBD | TBD | TBD | TBD |
| D-1500M | B2 | 0.35 | 2.871 | 2112.4 | 687991.5 | 12.74 | 34763 | 1.57 |
| D-1500M | D2 | 0.44 | 2.899 | 870.6 | 695946.5 | 4.76 | 11368 | 1.57 |
| D-1500M | P | 0.74 | 0.250 | 50418.0 | 50474.9 | 18.14 | 24856 | 1.57 |
| D-1500M | PD | 1.40 | 0.337 | 23901.5 | 71515.3 | 1.00 | 1626 | 1.57 |
| D-2500M | B0 | 7.89 | 0.129 | 13853.5 | 20049.7 | 12.03 | 3205 | 2.62 |
| D-2500M | B0S | TBD | TBD | TBD | TBD | TBD | TBD | TBD |
| D-2500M | B2 | 0.87 | 1.189 | 1402.1 | 278880.5 | 14.82 | 17428 | 2.62 |
| D-2500M | D2 | 1.25 | 1.014 | 515.1 | 236641.7 | 12.26 | 9860 | 2.62 |
| D-2500M | P | 2.51 | 0.104 | 14496.8 | 14548.9 | 4.45 | 3756 | 2.62 |
| D-2500M | PD | 3.21 | 0.166 | 8160.5 | 29773.9 | 2.51 | 2116 | 2.62 |

# moepagerd counters, PD @ 2500M:
{ "events": 11938, "tokens": 98, "prefetch_ops": 11938, "pin_ops": 0, "unpin_ops": 0, "demote_ops": 0, "failed_ops": 0, "budget_clamps": 0, "suppressed_prefetches": 0}
moepager_preload: MADV_RANDOM on 4018 MiB mapping (fd 3): ok
```

Figure 6. Collected end-to-end results (B0S rows are OOM-killed runs), daemon counters, and the shim at work.

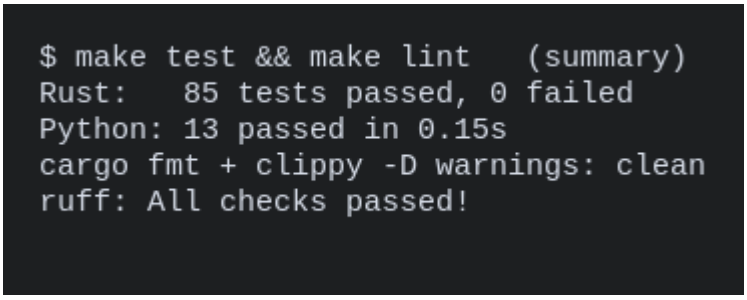


Figure 7. Test and lint status.

6. Results and Performance Evaluation

6.1 Setup

Item	Value
Machine	Laptop, Intel i5-12450H (8 cores / 12 threads, AVX2), 15 GiB RAM + 8 GiB zram swap, desktop session running
Storage	Solidigm P41 Plus 512 GB NVMe (PCIe 4, QLC); `/home` is btrfs with `compress=zstd:1`; btrfs readahead window 4 MiB
OS	Fedora 42, Linux 6.19.14, MGLRU enabled
Engine	llama.cpp [1] commit 970b51c (2026-10-22), CPU only, 8 threads
Model	OLMoE-1B-7B-0924 [8] Q4_K_M (4.21 GB; 16 layers × 64 experts, top-8; experts 3.90 GB, units ≈ 3.8 MB)
Memory budget	cgroup v2 `memory.max` = 1.5 GB or 2.5 GB, swap off unless stated; daemon and engine in the same cgroup
Cold start	model evicted from the page cache before every run

All numbers below are measured. Raw outputs are in `bench/results/phase7-olmoe/` (logs, CSVs, cgroup statistics, journal excerpts).

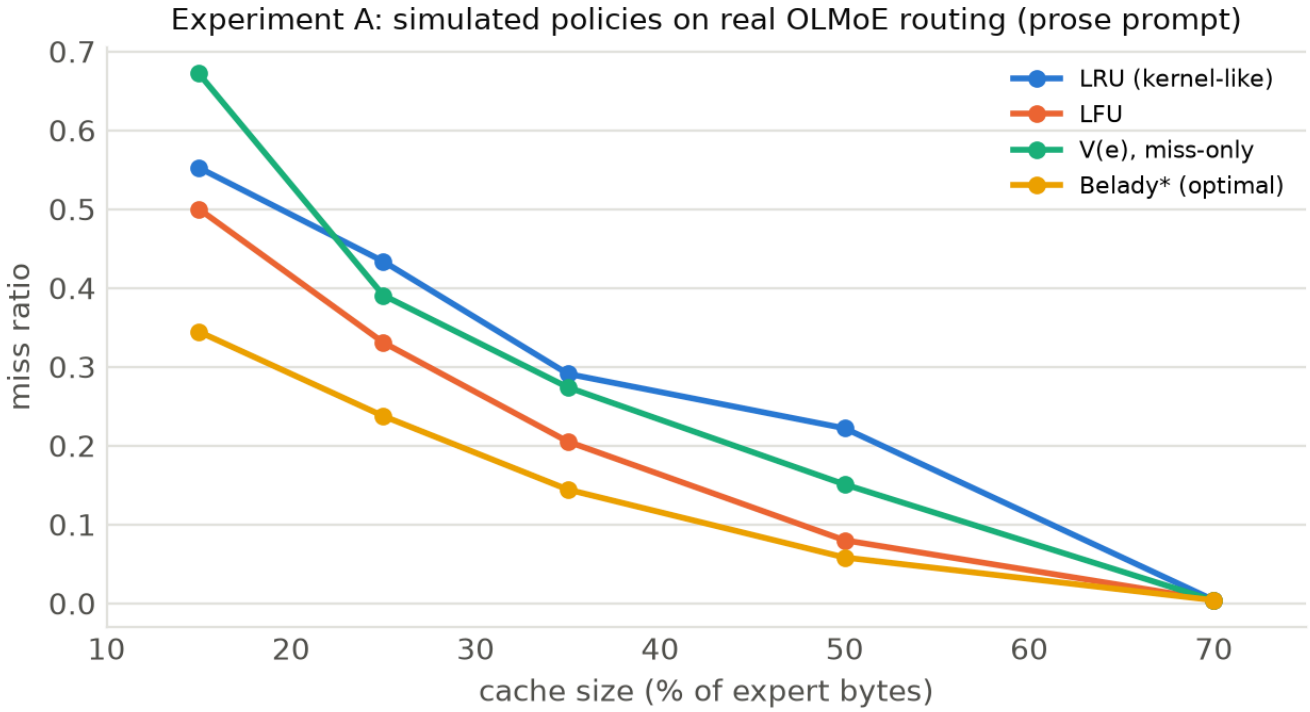
6.2 Experiment A: real routing, and how much any cache policy could gain

`moe_trace` recorded 255 decode tokens for each of three prompts: 32,640 expert uses per prompt. Measured properties:

Prompt	Same expert reused on the next token	Top-10 % experts' share of uses	Next-layer recall: transitions / popularity / previous token
chat	41 %	30 %	0.60 / 0.36 / 0.42
code	48 %	55 %	0.78 / 0.66 / 0.48
prose	45 %	38 %	0.69 / 0.45 / 0.46

Routing is strongly local in time (41–48 % reuse) and predictable across layers (60–78 % recall). The simulator was fed these traces with expert caches of 15–70 % of the expert bytes. Miss ratios for the prose prompt (other prompts in the CSVs):

Cache size	LRU	LFU	V(e), miss-only	Belady* (optimal)	Belady* saves vs LRU
15 %	0.552	0.500	0.672	0.345	38 %
25 %	0.434	0.331	0.391	0.237	45 %
35 %	0.291	0.205	0.274	0.144	50 %
50 %	0.222	0.080	0.151	0.058	74 %
70 %	0.004	0.004	0.004	0.004	4 %



Reading:

- An optimal policy would avoid 34–74 % of LRU's misses at 15–50 % capacity across all three prompts, so kill criterion **K1 does not fire**.
- Our V(e) residency policy captures only part of that gap. It beats LRU at 25–50 % but is *worse* at 15 %.
- Plain LFU beats V(e) in most cells.

Why V(e) lagged (pin-fraction sweep). The default lets V(e) pin only 60 % of the cache and leaves the rest to LRU. With full observation and a pin fraction of 0.95, V(e) matches LFU almost exactly: prose at 50 %, 0.081 vs 0.080; code at 15 %, 0.278 vs 0.276. Its ranking is sound.

The remaining gap is the **cost of black-box observation**. With misses only, V(e) at 0.95 has 10–40 % more misses than with full observation in most cells, and up to 2× in a few code-prompt cells (code at 25 %: 0.162 vs 0.075). That is exactly the information hidden from an unprivileged daemon, because page-cache hits are invisible.

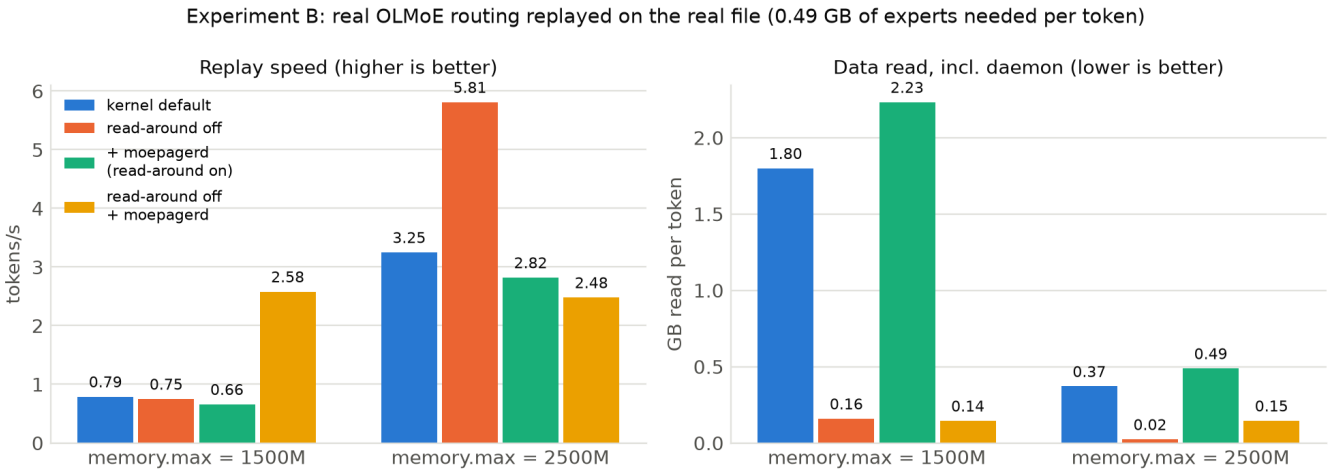
Pinning needs `CAP_IPC_LOCK`, so it was not used in the end-to-end runs. Data: `bench/results/phase7-olmoe/pinfrac-sweep/`.

6.3 Experiment B: what the real kernel does (and simulator validation)

The prose trace was replayed on the real model file through `mmap`, with 4 threads touching each expert's pages, under the memory wall. The trace needs 0.49 GB of expert data per token.

"Data read" is the page-cache inflow charged to the run's cgroup: reclaimed pages (`pgsteal`) plus file pages still charged at exit (Section 4.9). It includes the reads moepagerd issues on the engine's behalf. An earlier version of this table used the replayer's own `/proc/self/io`, which leaves out the daemon's reads and so understated traffic for the two daemon variants; on the four daemon-free runs the two counters agree to within 0.001 GB/token.

Budget	Variant	tokens/s	Data read / token (incl. daemon)	Major faults / token
1.5 GB	kernel default	0.79	**1.80 GB**	956
1.5 GB	read-around off (`MADV_RANDOM`)	0.75	**0.16 GB**	38,802
1.5 GB	+ moepagerd (read-around on)	0.66	2.23 GB	881
1.5 GB	**read-around off + moepagerd**	**2.58**	0.15 GB	12,342
2.5 GB	kernel default	3.25	0.37 GB	338
2.5 GB	read-around off	**5.81**	0.02 GB	5,704
2.5 GB	+ moepagerd (read-around on)	2.82	0.49 GB	321
2.5 GB	read-around off + moepagerd	2.48	0.15 GB	12,641



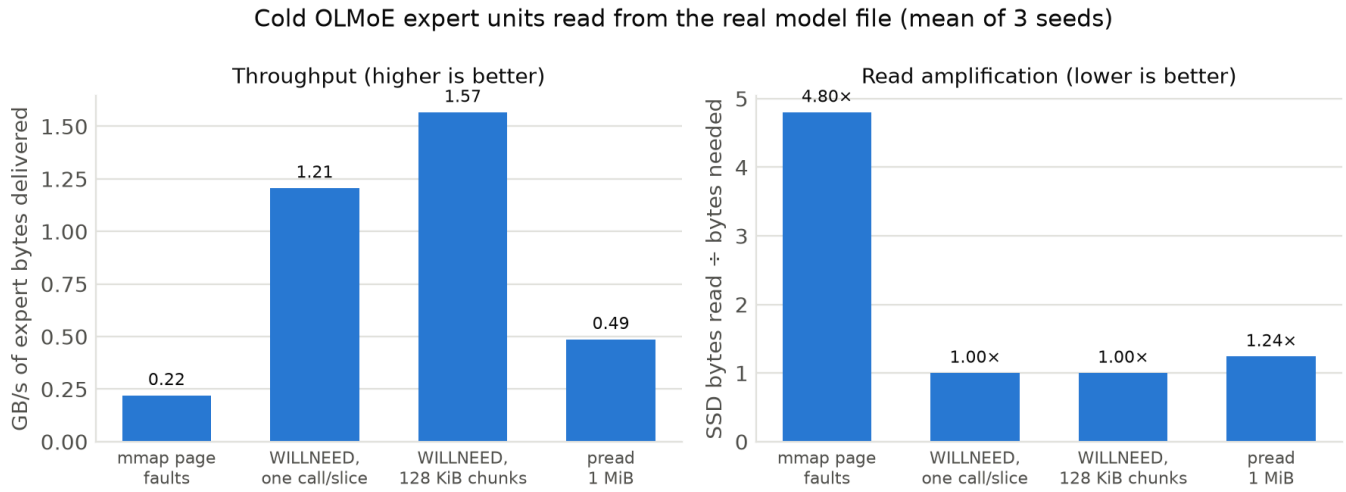
Reading:

- **Read-around pollution dominates.** With the kernel's default 4 MiB read-around, the SSD read 1.80 GB per token at 1.5 GB to deliver 0.49 GB, and nearly every expert access missed: neighbouring, unused experts filled the cache.
- **The simulator is validated once pollution is removed.** With read-around off, traffic fell 11× to 0.16 GB/token, which matches the simulator's LRU prediction for this capacity (0.14 GB/token, Section 6.2).
- **Fewer bytes alone is not faster.** Read-around off produced 39,000 small 4 KiB faults per token, so speed stayed at 0.75 tok/s.
- **moepagerd needs read-around off.** With read-around on, its sentinels fire on polluted pages and it made things slightly worse. With read-around off it turns the small faults into bulk reads: **3.3× faster** than the kernel default at 1.5 GB.
- At 2.5 GB, where most accesses already hit, read-around off alone was best in this replay; the daemon added overhead.

6.4 Experiment C: fault-driven vs bulk read bandwidth

32 cold expert units (3.8 MB each, three slices) of the real model file were read in three ways, 3 seeds each:

Method	GB/s delivered	SSD bytes read ÷ needed
mmap page faults (8 threads)	0.15–0.24	4.8×
WILLNEED, one call per slice	0.97–1.42	1.00×
WILLNEED, 128 KiB chunks	**1.49–1.70**	**1.00×**
pread, 1 MiB blocks	0.44–0.56	1.24×

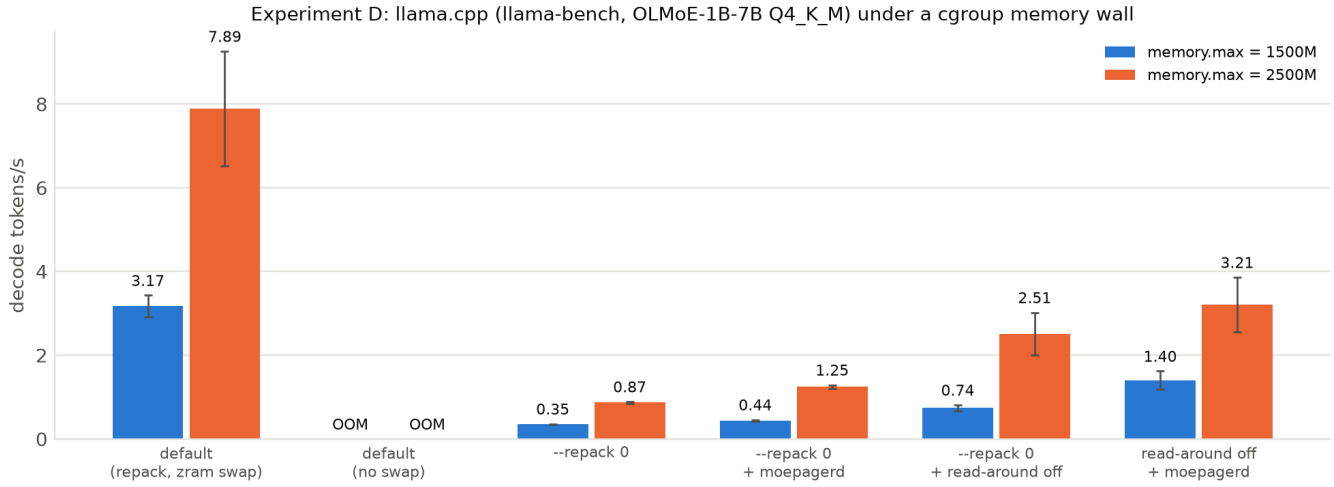


Bulk chunked readahead delivers expert bytes **≈ 7× faster** than page faults, with no amplification, so kill criterion **K2 does not fire**. Chunking matters: single calls are truncated by the kernel to the device readahead size.

6.5 Experiment D: end to end with llama.cpp

`llama-bench` decoded 32 tokens × 3 repetitions (after a 1-token warmup) inside the cgroup, starting cold. Decode throughput is mean ± standard deviation over the 3 repetitions. SSD traffic and major faults are cgroup/device totals divided by all 97 decoded tokens. PSI is the cgroup's memory-pressure "some" average over the last 10 s at exit.

Budget	Configuration	tokens/s	SSD GB / token	Major faults / token	Memory PSI (some, avg10)
1.5 GB	B0 default llama.cpp (repack; zram swap allowed)	3.17 ± 0.26	0.31	38,206	16.2 %
1.5 GB	B0S default llama.cpp, no swap	**OOM-killed**	—	—	—
1.5 GB	B2 `--repack 0` (page cache, kernel only)	0.35 ± 0.01	2.87	2,112	12.7 %
1.5 GB	D2 B2 + moepagerd	0.44 ± 0.02	2.90	871	4.8 %
1.5 GB	P B2 + read-around off (shim)	0.74 ± 0.07	0.25	50,418	18.1 %
1.5 GB	**PD B2 + read-around off + moepagerd**	**1.40 ± 0.22**	0.34	23,902	**1.0 %**
2.5 GB	B0 default llama.cpp (repack; zram swap allowed)	7.89 ± 1.37	0.13	13,854	12.0 %
2.5 GB	B0S default llama.cpp, no swap	**OOM-killed**	—	—	—
2.5 GB	B2 `--repack 0`	0.87 ± 0.02	1.19	1,402	14.8 %
2.5 GB	D2 B2 + moepagerd	1.25 ± 0.04	1.01	515	12.3 %
2.5 GB	P B2 + read-around off	2.51 ± 0.51	0.10	14,497	4.5 %
2.5 GB	**PD B2 + read-around off + moepagerd**	**3.21 ± 0.65**	0.17	8,161	2.5 %



Against the same engine serving weights from the page cache (B2):

- PD is **4.0× faster at 1.5 GB** and **3.7× faster at 2.5 GB**.
- PD reads **8.4× and 7.0× fewer bytes from the SSD** per token.
- PD cuts memory-pressure stall (PSI) from 12.7 % to 1.0 % at 1.5 GB. This matters for other applications on the machine.

Splitting the gain:

- Turning read-around off accounts for 2.1–2.9×.

- moepagerd's completion readahead adds a further 1.9× (1.5 GB) and 1.3× (2.5 GB) on top.
- moepagerd *without* read-around off (D2) helps only 1.3–1.4×.

Variability: PD at 2.5 GB was measured in three independent sessions: 3.30 (1 repetition), 5.42 ± 1.07 and 3.21 ± 0.65 tok/s. The table reports the last.

Against default llama.cpp (B0):

- B0 is faster (3.17 and 7.89 tok/s), but it gets there by pushing **2.5 GB and 2.3 GB of repacked weights into zram**, compressed swap that lives in RAM and is not counted against `memory.max` (journal: "1.4G memory peak, 2.5G memory swap peak"). It therefore used far more than the budget.
- With swap disabled, so it is held to the same budget (B0S), default llama.cpp was **OOM-killed at both budgets**: it cannot run at all.
- moepager's configuration runs inside the budget.

A fair speed comparison against default llama.cpp therefore needs a machine whose swap is on disk, or a budget that includes zram. That is future work.

Ablation (refractory window): suppressing repeated completion requests for 20 ms looked like an obvious efficiency fix, but it doubled misses and cut PD at 2.5 GB from 3.21–5.42 to 1.09–2.05 tok/s. The repeats re-fetch pages reclaimed while the engine is still working through an expert. The window is therefore off by default.

6.6 Second model: gpt-oss-20b (experiments A–C)

The same pipeline was run on gpt-oss-20b MXFP4: 12.1 GB file, 24 layers × 32 experts, top-4, 13.25 MB per expert unit. Memory limits were 4 GB and 6 GB, and routing was recorded for 3 prompts × 128 tokens. Raw data: `bench/results/phase7-gptoss/`.

Experiment D did not run. Its first configuration, default llama.cpp, repacks 100 % of gpt-oss's expert bytes (≈ 10 GB) into anonymous memory and swaps them to zram, which lives in RAM. The machine ran critically low on memory and the run was stopped. This is itself evidence of the problem moepager avoids, but there are no end-to-end gpt-oss numbers.

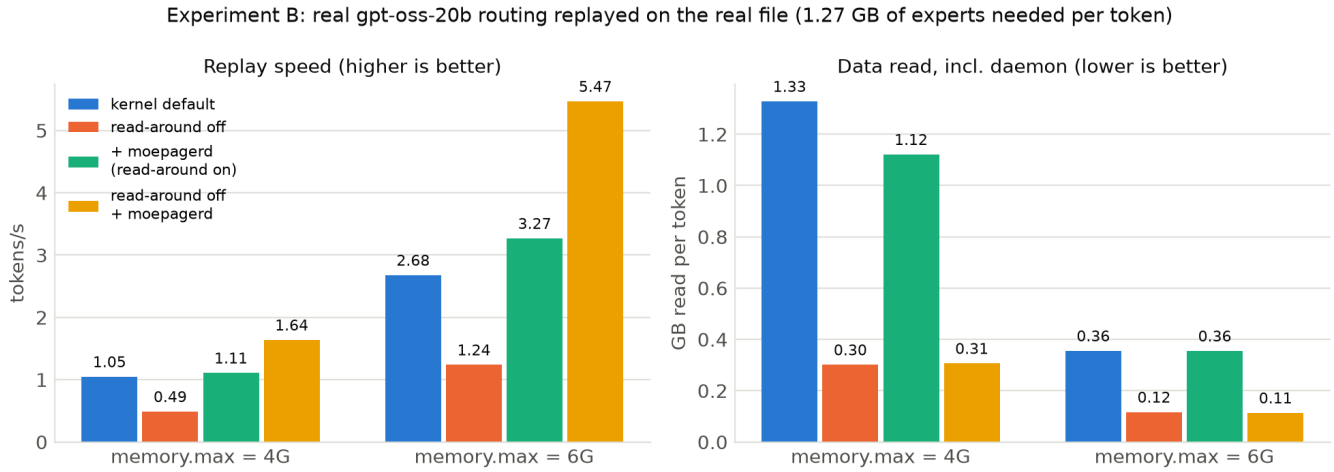
- **Routing (A):** 40–47 % token-to-token reuse, next-layer recall 0.54–0.62 versus 0.39–0.46 for popularity. Belady* saves 44–67 % of LRU's misses at 15–50 % capacity, similar to OLMoE.

- **Bandwidth (C), 13.25 MB units:**

Method	GB/s	Amplification
mmap faults	0.14–0.44	≈ 2.5×
WILLNEED, 128 KiB chunks	1.44–1.88	1.00×
WILLNEED, one call per slice	0.93–1.10	1.44×, truncated
pread	0.31–0.65	—

- **Replay (B), prose trace, 1.27 GB of experts per token:**

Budget	Variant	tokens/s	Data read / token (incl. daemon)
4 GB	kernel default	1.05	1.33 GB
4 GB	read-around off	0.49	0.30 GB
4 GB	+ moepagerd (read-around on)	1.11	1.12 GB
4 GB	**read-around off + moepagerd**	**1.64**	**0.31 GB**
6 GB	kernel default	2.68	0.36 GB
6 GB	read-around off	1.24	0.12 GB
6 GB	+ moepagerd (read-around on)	3.27	0.36 GB
6 GB	**read-around off + moepagerd**	**5.47**	**0.11 GB**



Reading:

- The best configuration is the same as for OLMoE: read-around off plus completion readahead, **1.6× and 2.0× faster** than the kernel default, with **3–4× less data read** (0.31 vs 1.33 and 0.11 vs 0.36 GB/token).
- Two differences from OLMoE:
  - gpt-oss's expert slices (4.4 MB) are about as large as the read-around window, so read-around wastes less (≈ 1.05× at 4 GB).
  - Read-around off *alone* is slower than the default here, because 4 KiB faults replace useful large reads.
- moepagerd turns those small faults back into bulk reads, and also helps somewhat on its own with read-around on (1.06× and 1.22×).

6.7 Comparison with expected outcomes and with existing methods

Expectation (IDEA_REVIEW, before measuring)	Outcome
Smarter eviction/pinning is the main lever (≥ 15 % fewer misses possible, K1)	The gap exists (Belady* saves 34–74 % vs LRU), but our online V(e) policy did not capture it reliably, and pinning needs privileges. **Not the main lever.**
Bulk readahead beats fault-driven reads (K2)	Confirmed: ≈ 7× (Exp. C).
Default llama.cpp is beaten (K3)	Only at equal memory, where it OOMs. With zram it is faster. **Partially.**
(not anticipated)	Fault read-around pollution is the dominant cost: 3.7–4.8× read amplification. Turning it off is required for both correctness of black-box detection and speed.

Relation to existing methods:

- Si et al. [20] found that `fadvise` lookahead hints gain ≤ 5 % on a GH200 with a `pread`-based engine. Our result differs because on an mmap-based engine the problem is *how* missing experts are read (small synchronous faults plus large wasted read-around), not *which* experts to keep. Completion readahead attacks exactly that.
- In-engine systems (MoE-Infinity [14], HOBBIT [15], llama.cpp PR #25294 [19]) see router decisions directly and can avoid the issue entirely. moepager reaches a 3.7–4.0× improvement over the stock page-cache path without modifying the engine's code or binary, at the cost of one `LD_PRELOAD` hint.

6.8 Go / no-go decision

GO, with narrowed scope (recorded in PHASES.md):

- The product is "read-around off + expert-completion readahead", validated end to end.
- V(e) pinning and predictive prefetch are demoted to research options.
- The next phase must validate on a second machine with disk swap and ext4, and on a larger model.

7. Conclusion

7.1 Summary

We built moepager, a toolchain and daemon that manages the Linux page cache for Mixture-of-Experts inference using the model file's structure, without changing the inference engine's code.

The project was run as an experiment with kill criteria fixed in advance. On a real model (OLMoE-1B-7B) on a 15 GB laptop, the outcome was:

- **Go on the mechanism, no-go on the original emphasis.** We expected smarter *eviction* to be the main lever. Measurement showed the dominant cost under memory pressure is the kernel's fault **read-around**. On this machine's btrfs it read 3.7× more data than needed and filled the cache with unused neighbouring experts.

- **What works:**

    - turning read-around off for the model mapping, with a 64-line `LD_PRELOAD` shim, since no stock llama.cpp option does this on single-node machines;
    - plus moepagerd's expert-completion readahead in 128 KiB chunks.

    Together they turn tens of thousands of 4 KiB faults per token into bulk reads. See Section 6.5 for the end-to-end numbers. - **What doesn't (yet):** - Completion readahead *without* turning read-around off made things worse. - Our V(e) pinning policy only matched LFU when allowed to pin most of the cache. With black-box (miss-only) observation it still loses 10–40 % in most cells, up to 2×, and it needs extra privileges. - Default llama.cpp is faster than every page-cache configuration *if* it may spill 2.3–2.5 GB of repacked weights into zram (compressed RAM). Without swap it is OOM-killed at both budgets. - **Smaller findings with independent value:** - The kernel silently truncates `WILLNEED`. - llama.cpp's default repacking moves 77–100 % of expert bytes out of the page cache. - llama.cpp's `--lazy-mode` keeps weights in anonymous memory. - btrfs reports different device numbers in `stat()` and `/proc/pid/maps`.

7.2 Limitations

- One machine, one filesystem (btrfs with a 4 MiB readahead window). End-to-end llama.cpp numbers exist only for OLMoE. gpt-oss-20b was evaluated with real routing, replay and the bandwidth benchmark, but its llama-bench run was stopped for low system memory. Qwen3-30B was only mapped from its header.
- Short runs (32 decode tokens × 3 repetitions in llama-bench); some configurations show large run-to-run spread.
- A desktop session was running during measurements.
- Pinning (`mlock`), demotion (`MADV_COLD`) and eBPF observation are implemented but untested, because they need privileges.
- The shim is still an intervention in the engine's process, although it doesn't change its code or binary.

7.3 Future work

1. Upstream: propose a llama.cpp option to apply `MADV_RANDOM` to expert tensors on single-node machines, which would remove the need for the shim.
2. Repeat on ext4 (128 KiB readahead) and on the larger models, with longer runs.
3. Evaluate pinning end to end with `CAP_IPC_LOCK` and a pin budget near 90 % of the expert cache. Narrow the miss-only gap with privileged hit observation (DAMON [23]) where available.
4. Add a fallback when WILLNEED is ineffective (`MADV_POPULATE_READ`), and PSI-driven [29] control of the memory budget to protect other applications.

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